

Lorem Ipsum Dolor Sit Amet

Lorem Ipsum

Dolor Sit

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Lorem Ipsum University

Abstract

The task of motion planning is equivariant to rigid-body transformations, but the field the flow matching network learns is not. There are currently three techniques to restore equivariance: in the data, in the operator, or in the weights.

We ask what the task gives us before any of them. The task itself already supplies us with a frame in the workspace, by placing the origin at the midpoint of the start and goal and aligning the first axis with the direction between the two points, which removes five of the six degrees of freedom of $SE(3)$ exactly, at initialisation and at no cost. On 3D workspaces holding 40 obstacles, this raises the held-out collision-free rate from 14.6% to 51.1%. A straight line scores 15.6% on the identical problems, and the world-frame model does not beat it. A control that shows the obstacle encoder the start and goal without changing frame also reaches 15.6%, so it is the frame and not the conditioning that moves the model off the floor. One degree of freedom survives, a roll about that axis, and no continuous rule can fix it. Augmentation, frame averaging and an $SO(2)$ -equivariant architecture are worth +0.8, +0.68 and +0.15 points, and the constrained backbone reaches any level two to three times sooner, so the symmetry buys optimisation rather than accuracy. The frame is also not the largest effect available: a per-waypoint signed distance and gradient is worth +40.0 against the frame's +36.5, and the frame still adds +24.8 alongside it.

1 Introduction

Motion planning is a fundamental problem in robotics, requiring the generation of collision-free, kinematically feasible trajectories from a start to a goal configuration in complex environments. Equivariance to rigid-body transformations is a cornerstone of generalization in this setting. It ensures that a planner solves structurally identical tasks identically, regardless of where in the world they are posed. Sampling-based [14] and optimization-based [30, 24] algorithms are equivariant by construction, since they act on the geometry directly, and they offer completeness or convergence guarantees. Their per-query computational cost limits real-time applicability. Learning-based planners [22, 8, 12, 3] amortize that cost across queries and have emerged as a promising alternative. They do not inherit the symmetry. A network trained on world coordinates relearns the same manoeuvre at every position and orientation.

Existing work restores it in one of three parts of the pipeline. In the data, through augmentation. In the inference operator, through frame averaging or canonicalization [20, 13]. Or in the weights, through architectures equivariant by construction [5, 26, 9, 6], as adopted in recent robot learning systems [25, 23, 28, 29]. All three modify the model. None measures how much of the symmetry the problem statement already supplies before any model is built.

A planning query names a start and a goal, and those two points determine a frame. Place the origin at their midpoint, align the first axis with the direction between them, and express every waypoint and every obstacle in that frame. Three translational and two rotational degrees of freedom vanish exactly, at initialisation, at no computational cost and with no constraint on the architecture. The six numbers that conditioned the network on the query collapse to one invariant scalar. The frame needs a second axis to be determined, and ours takes it from a fixed world axis, so rotating the whole problem rolls the frame rather than leaving it unchanged. What survives is exactly that roll: $SE(3)$ collapses to $SO(2)$, not to nothing. No continuous rule can fix the remainder [17], so the sixth degree of freedom must be handled by one of the three mechanisms above. Fig. 1 takes the group apart one stage at a time.

We build all three on a cluttered 3D benchmark, hold architecture, data and training budget fixed, and calibrate against a trivial floor and four classical planners on the identical held-out problems. The frame is worth +36.5 points of held-out collision-free rate. The three mechanisms for the sixth are worth +0.8, +0.68 and +0.15. A straight line from start to goal scores 15.6%, and the world-frame model, trained to a plateau, reaches 14.60%.

Our contributions are as follows.

- **Five of six degrees of freedom are free.** A frame derived from the query in closed form removes them exactly, raising the held-out collision-free rate from $14.60 \pm 0.20\%$ to $51.10 \pm 0.25\%$ across three seeds, with no change to the architecture, the data or the budget.
- **All three mechanisms for the sixth, measured against each other.** None recovers a point. We explain that rather than inferring it from flat curves, by measuring the non-equivariance residual of the learned field and proving it bounds what any post-hoc symmetrisation can remove. We then show the residual does not predict what a mechanism is

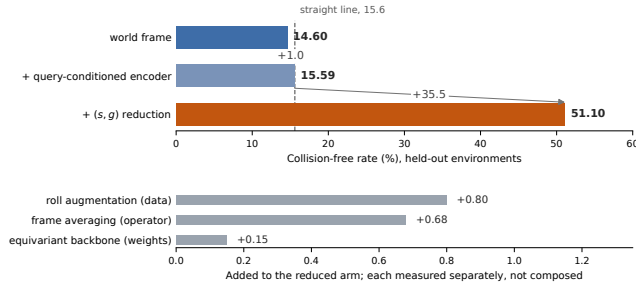


Figure 1: **Sixty environments, trained to convergence, three seeds**, the setting used throughout this paper. *Top*: a frame read off the query in closed form is worth +35.5 points, and showing the obstacle encoder the same query without changing frame is worth +1.0. The dashed line is a straight segment from start to goal on the identical problems, which the world-frame arm does not beat. *Bottom*: the three mechanisms for the one degree of freedom the frame cannot remove. Each is measured against its own baseline and they are alternatives rather than a decomposition, so they are not composed here and were never composed in the experiments.

worth: two arms that are equally equivariant differ by 28 points.

- **Calibration in both directions, including against ourselves.** Supplying each waypoint with local geometry is worth +40.0 points where the frame is worth +36.5, so the representation change studied here is not the largest effect available on this benchmark. It is not a substitute for that information either: with local geometry present, the frame still adds +24.8.

2 Related Work

The task of motion planning is equivariant to rigid-body transformations, but the field the flow matching network learns is not. There are currently three techniques to restore equivariance: in the data, in the operator, or in the weights.

We ask what the task gives us before any of them. The task itself already supplies us with a frame in the workspace, by placing the origin at the midpoint of the start and goal and aligning the first axis with the direction between the two points, which removes five of the six degrees of freedom of $SE(3)$ exactly, at initialisation and at no cost. On 3D workspaces holding 40 obstacles, this raises the held-out collision-free rate from 14.6% to 51.1%. A straight line scores 15.6% on the identical problems, and the world-frame model does not beat it. A control that shows the obstacle encoder the start and goal without changing frame also reaches 15.6%, so it is the frame and not the conditioning that moves the

model off the floor. One degree of freedom survives, a roll about that axis, and no continuous rule can fix it. Augmentation, frame averaging and an $SO(2)$ -equivariant architecture are worth +0.8, +0.68 and +0.15 points, and the constrained backbone reaches any level two to three times sooner, so the symmetry buys optimisation rather than accuracy. The frame is also not the largest effect available: a per-waypoint signed distance and gradient is worth +40.0 against the frame’s +36.5, and the frame still adds +24.8 alongside it.

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3 Method

3.1 Decomposing the group

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Proposition 1 (Lorem ipsum). *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua $X' = R_{\hat{x}}(\theta) X$.*

Proposition 2 (Dolor sit amet). *Lorem ipsum dolor sit amet consectetur adipiscing elit sed do eiusmod tempor incididunt ut labore.*

Proof. Lorem ipsum dolor sit amet consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua ut enim ad minim veniam. $\square$

4 Setup and Calibration

4.1 Benchmark and protocol

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lorem ipsum	00.0	0.000	<0.001
dolor sit	00.0	0.000	0.000
amet consectetur	00.0	0.000	0.000
adipiscing elit	00.0	0.000	0.000
sed do eiusmod	00.0	0.000	0.000
tempor incididunt	00.0	−0.000	0.000
ut labore et dolore	00.0	−0.000	0.000

Table 1: Lorem ipsum dolor sit amet consectetur adipiscing elit sed do eiusmod tempor incididunt ut labore et dolore magna aliqua

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4.2 The floor, and a paired test against it

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5 Results

5.1 Scaling

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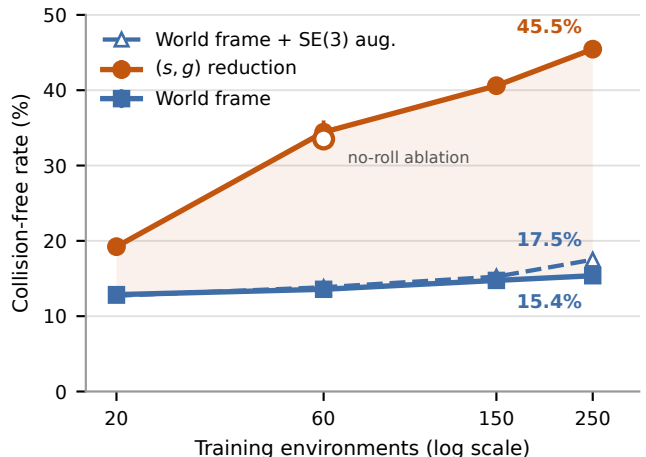


Figure 2: Lorem ipsum dolor sit amet consectetur adipiscing elit

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5.2 Control: a query-conditioned encoder

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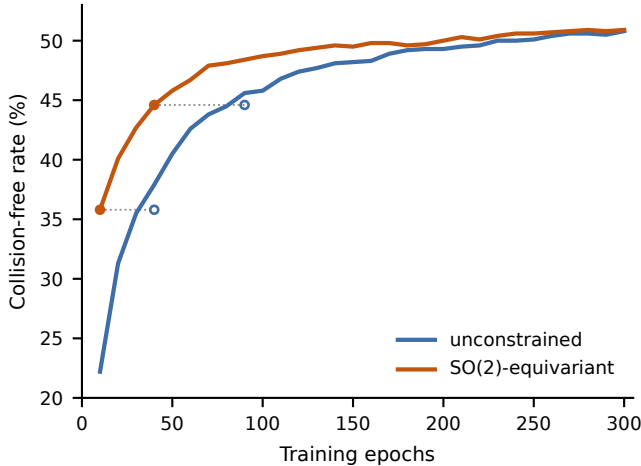


Figure 3: Lorem ipsum dolor sit amet consectetur adipiscing

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5.3 Control: local geometry, and how it scales

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obstacles, this raises the held-out collision-free rate from 14.6% to 51.1%. A straight line scores 15.6% on the identical problems, and the world-frame model does not beat it. A control that shows the obstacle encoder the start and goal without changing frame also reaches 15.6%, so it is the frame and not the conditioning that moves the model off the floor. One degree of freedom survives, a roll about that axis, and no continuous rule can fix it. Augmentation, frame averaging and an $SO(2)$ -equivariant architecture are worth +0.8, +0.68 and +0.15 points, and the constrained backbone reaches any level two to three times sooner, so the symmetry buys optimisation rather than accuracy. The frame is also not the largest effect available: a per-waypoint signed distance and gradient is worth +40.0 against the frame’s +36.5, and the frame still adds +24.8 alongside it.

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5.4 Replications

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available: a per-waypoint signed distance and gradient is worth +40.0 against the frame’s +36.5, and the frame still adds +24.8 alongside it.

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5.5 The residual, and what it bounds

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6 Limitations

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7 Conclusion

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Acknowledgements

The authors thank the IWIN-FINS Laboratory at Shanghai Jiao Tong University for the computational resources used in this work.

The software implementation was carried out with Claude Code under the first author’s direction. The implementations of the classical planners used to generate expert trajectories follow the published algorithms [14, 30, 24] and their public reference implementations. The problem formulation, the theoretical analysis, the experimental design and the text of this paper are the authors’ own. Every reported number is produced by the released code, whose geometric and equivariance-critical components are verified by property-based tests that run on untrained networks and include negative controls.

References

- [1] M. S. Albergo and E. Vanden-Eijnden. Building normalizing flows with stochastic interpolants. In *ICLR*, 2023.

- [2] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv:2104.13478*, 2021.
- [3] J. Carvalho, A. T. Le, M. Baierl, D. Koert, and J. Peters. Motion planning diffusion: Learning and planning of robot motions with diffusion models. In *IROS*, 2023.
- [4] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *RSS*, 2023.
- [5] T. S. Cohen and M. Welling. Group equivariant convolutional networks. In *ICML*, 2016.
- [6] C. Deng, O. Litany, Y. Duan, A. Poulenard, A. Tagliasacchi, and L. Guibas. Vector neurons: A general framework for $SO(3)$ -equivariant networks. In *ICCV*, 2021.
- [7] B. Elesedy and S. Zaidi. Provably strict generalisation benefit for equivariant models. In *ICML*, 2021.
- [8] A. Fishman, A. Murali, C. Eppner, B. Peele, B. Boots, and D. Fox. Motion policy networks. In *CoRL*, 2022.
- [9] F. B. Fuchs, D. E. Worrall, V. Fischer, and M. Welling. SE(3)-transformers: 3D roto-translation equivariant attention networks. In *NeurIPS*, 2020.
- [10] M. Geiger and T. Smidt. e3nn: Euclidean neural networks. *arXiv:2207.09453*, 2022.
- [11] N. Gruver, M. Finzi, M. Goldblum, and A. G. Wilson. The Lie derivative for measuring learned equivariance. In *ICLR*, 2023.
- [12] M. Janner, Y. Du, J. B. Tenenbaum, and S. Levine. Planning with diffusion for flexible behavior synthesis. In *ICML*, 2022.
- [13] S.-O. Kaba, A. K. Mondal, Y. Zhang, Y. Bengio, and S. Ravanbakhsh. Equivariance with learned canonicalization functions. In *ICML*, 2023.
- [14] J. J. Kuffner and S. M. LaValle. RRT-connect: An efficient approach to single-query path planning. In *ICRA*, 2000.
- [15] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. In *ICLR*, 2023.
- [16] X. Liu, C. Gong, and Q. Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *ICLR*, 2023.
- [17] J. Milnor. Analytic proofs of the “hairy ball theorem” and the Brouwer fixed point theorem. *American Mathematical Monthly*, 85(7):521–524, 1978.
- [18] A. van den Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, A. Graves, N. Kalchbrenner, A. Senior, and K. Kavukcuoglu. WaveNet: A generative model for raw audio. *arXiv:1609.03499*, 2016.
- [19] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville. FiLM: Visual reasoning with a general conditioning layer. In *AAAI*, 2018.
- [20] O. Puny, M. Atzmon, H. Ben-Hamu, I. Misra, A. Grover, E. J. Smith, and Y. Lipman. Frame averaging for invariant and equivariant network design. In *ICLR*, 2022.
- [21] C. R. Qi, H. Su, K. Mo, and L. J. Guibas. PointNet: Deep learning on point sets for 3D classification and segmentation. In *CVPR*, 2017.
- [22] A. H. Qureshi, A. Simeonov, M. J. Bency, and M. C. Yip. Motion planning networks. In *ICRA*, 2019.
- [23] H. Ryu, H.-i. Lee, J.-H. Lee, and J. Choi. Equivariant descriptor fields: SE(3)-equivariant energy-based models for end-to-end visual robotic manipulation learning. In *ICLR*, 2023.
- [24] J. Schulman, Y. Duan, J. Ho, A. Lee, I. Awwal, H. Bradlow, J. Pan, S. Patil, K. Goldberg, and P. Abbeel. Motion planning with sequential convex optimization and convex collision checking. *IJRR*, 33(9):1251–1270, 2014.
- [25] A. Simeonov, Y. Du, A. Tagliasacchi, J. B. Tenenbaum, A. Rodriguez, P. Agrawal, and V. Sitzmann. Neural descriptor fields: SE(3)-equivariant object representations for manipulation. In *ICRA*, 2022.
- [26] N. Thomas, T. Smidt, S. Kearnes, L. Yang, L. Li, K. Kohlhoff, and P. Riley. Tensor field networks: Rotation- and translation-equivariant neural networks for 3D point clouds. *arXiv:1802.08219*, 2018.
- [27] J. Urain, N. Funk, J. Peters, and G. Chelvatzaki. SE(3)-DiffusionFields: Learning smooth cost functions for joint grasp and motion optimization through diffusion. In *ICRA*, 2023.
- [28] D. Wang, S. Hart, D. Surovik, T. Kelestemur, H. Huang, H. Zhao, M. Yeatman, J. Wang, R. Walters, and R. Platt. Equivariant diffusion policy. In *CoRL*, 2024.
- [29] J. Yang, Z. Cao, C. Deng, R. Antonova, S. Song, and J. Bohg. EquiBot: SIM(3)-equivariant diffusion policy for generalizable and data efficient learning. In *CoRL*, 2024.
- [30] M. Zucker, N. Ratliff, A. D. Dragan, M. Pivtoraiko, M. Klingensmith, C. M. Dellin, J. A. Bagnell, and S. S. Srinivasa. CHOMP: Covariant Hamiltonian optimization for motion planning. *IJRR*, 32(9-10):1164–1193, 2013.